

Brutal Practice Paper B48 - Class 7 (2x Hard, Fusion): Nutrition in Animals, Motion & Time, Fractions & Decimals, The Triangle & its Properties

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. In the small intestine, finger-like villi soak up glucose. One villus absorbs 0.6 mg of glucose each hour. A patch has 2,500 working villi. Working for 3 hours, how many grams of glucose does the patch absorb?

[Nutrition in Animals]

- (A) 4.5 g
- (B) 4500 mg
- (C) 0.45 g
- (D) 13.5 g
- (E) 1.5 g
- (F) 45 g

2. Work out $(\frac{2}{3} + \frac{3}{4})$ divided by $(\frac{5}{6} - \frac{1}{4})$ and give the answer as a fraction in lowest terms. [Fractions &

Decimals]

- (A) $\frac{24}{35}$
- (B) $\frac{10}{7}$
- (C) $\frac{7}{17}$
- (D) $\frac{5}{7}$
- (E) $\frac{17}{5}$
- (F) $\frac{17}{7}$
- (G) $\frac{17}{12}$

3. A train covers 180 km in 2 hours 30 minutes at a steady pace. What is its speed in metres per second?

[Motion & Time]

- (A) 12 m/s
- (B) 20 m/s
- (C) 18 m/s
- (D) 36 m/s
- (E) 25 m/s
- (F) 72 m/s
- (G) 50 m/s

4. Inside a triangle the three angles have the ratio 3 : 5 : 7. Work out the measure of its biggest angle. [The

Triangle & its Properties]

- (A) 60 degrees
- (B) 84 degrees
- (C) 48 degrees
- (D) 90 degrees
- (E) 100 degrees
- (F) 72 degrees
- (G) 36 degrees

5. A buffalo swallows 30 kg of grass quickly. Two-fifths of it is stored as a soft mass, and later three-quarters of that stored mass is brought back up as cud to chew again. How many kilograms come back as cud? [\[Nutrition in Animals\]](#)

- (A) 9 kg
- (B) 4.5 kg
- (C) 18 kg
- (D) 12 kg
- (E) 22.5 kg
- (F) 6 kg

6. A water tank is $\frac{3}{5}$ full. After 24 litres are poured in, it becomes $\frac{4}{5}$ full. What is the full capacity of the tank in litres? [\[Fractions & Decimals\]](#)

- (A) 120 L
- (B) 96 L
- (C) 24 L
- (D) 72 L
- (E) 40 L
- (F) 150 L

7. A car travels the first 60 km at 40 km/h and the next 60 km at 60 km/h. What is its average speed for the whole 120 km trip? [\[Motion & Time\]](#)

- (A) 50 km/h
- (B) 55 km/h
- (C) 48 km/h
- (D) 52 km/h
- (E) 100 km/h
- (F) 24 km/h
- (G) 45 km/h

8. An exterior angle of a triangle measures 125 degrees, and one of its two remote interior angles is 47 degrees. What is the other remote interior angle? [\[The Triangle & its Properties\]](#)

- (A) 78 degrees
- (B) 47 degrees
- (C) 72 degrees
- (D) 35 degrees
- (E) 102 degrees
- (F) 55 degrees
- (G) 125 degrees

9. A meal contains 80 g of protein. In the stomach 35% of it is broken down. In the small intestine, 60% of the protein still left is then digested. How many grams of protein remain undigested at the end? [\[Nutrition in Animals\]](#)

- (A) 48 g
- (B) 20.8 g
- (C) 20 g
- (D) 31.2 g
- (E) 28 g
- (F) 8.4 g
- (G) 52 g

10. Evaluate $1.25 \times 0.8 + 3.6 \div 0.9 - 2.5$, following the correct order of operations. [\[Fractions & Decimals\]](#)

- (A) 0.5
- (B) 5.0
- (C) 2.5
- (D) 3.5
- (E) 2.0
- (F) 4.5
- (G) 1.5

11. A simple pendulum makes 30 complete swings in 48 seconds. What is its time period (the time for one swing)? [\[Motion & Time\]](#)

- (A) 3.2 s
- (B) 0.8 s
- (C) 48 s
- (D) 2 s
- (E) 0.625 s
- (F) 1.6 s
- (G) 1.5 s

12. A right-angled triangle has its two shorter sides (legs) measuring 9 cm and 12 cm. What is the length of its hypotenuse? [\[The Triangle & its Properties\]](#)

- (A) 15 cm
- (B) 225 cm
- (C) 13 cm
- (D) 18 cm
- (E) 10.5 cm
- (F) 21 cm

13. An Amoeba traps food by pushing out false feet (pseudopodia). It captures 6 food particles every 5 minutes at a steady rate. How many particles does it trap in 2 hours 30 minutes? [\[Nutrition in Animals\]](#)

- (A) 30 particles
- (B) 180 particles
- (C) 150 particles
- (D) 36 particles
- (E) 72 particles
- (F) 300 particles

14. A recipe needs $2 \frac{3}{4}$ cups of flour for one batch. How many cups of flour are needed for $2 \frac{1}{2}$ batches? Give the answer as a mixed number. [\[Fractions & Decimals\]](#)

- (A) $5 \frac{1}{2}$ cups
- (B) 7 cups
- (C) $6 \frac{1}{4}$ cups
- (D) $5 \frac{1}{4}$ cups
- (E) $5 \frac{5}{8}$ cups
- (F) $6 \frac{7}{8}$ cups
- (G) $6 \frac{1}{2}$ cups

15. Sunlight takes 8 minutes 20 seconds to reach Earth. Taking the speed of light as 3×10^8 m/s, what is the distance from the Sun to Earth in kilometres? [\[Motion & Time\]](#)

- (A) 1.5×10^8 km
- (B) 3×10^8 km
- (C) 1.5×10^{11} km
- (D) 9×10^9 km
- (E) 1.5×10^9 km
- (F) 1.5×10^7 km
- (G) 5×10^8 km

16. An isosceles triangle has a vertex (top) angle of 40 degrees. What is the size of each of its two equal base angles? [\[The Triangle & its Properties\]](#)

- (A) 50 degrees
- (B) 80 degrees
- (C) 70 degrees
- (D) 40 degrees
- (E) 100 degrees
- (F) 140 degrees
- (G) 35 degrees

17. Bile emulsifies fat. A test shows 45 g of fat needs exactly 9 mL of bile. A person eats a fatty meal with 72 g of fat. At the same rate, how many millilitres of bile are needed? [\[Nutrition in Animals\]](#)

- (A) 14.4 mL
- (B) 1.44 mL
- (C) 16 mL
- (D) 144 mL
- (E) 14 mL
- (F) 12.8 mL

18. In the number 47.385, by how much does the place value of the digit 3 exceed the place value of the digit 8? [\[Fractions & Decimals\]](#)

- (A) 0.38
- (B) 2.2
- (C) 0.22
- (D) 0.05
- (E) 0.5
- (F) 0.3
- (G) 5

19. A car's speedometer shows a steady 90 km/h. How many metres does the car travel in 4 seconds? [\[Motion & Time\]](#)

- (A) 360 m
- (B) 75 m
- (C) 90 m
- (D) 400 m
- (E) 60 m
- (F) 100 m

20. Two sides of a triangle are 7 cm and 10 cm. For how many different whole-number values of the third side is a triangle possible? [\[The Triangle & its Properties\]](#)

- (A) 3 values
- (B) 16 values
- (C) 12 values
- (D) 13 values
- (E) 17 values
- (F) 14 values
- (G) 15 values

21. An energy bar has 45 g of carbohydrate (4 kcal per gram) and 20 g of fat (9 kcal per gram). What percent of the bar's energy comes from the fat? [\[Nutrition in Animals\]](#)

- (A) 25%
- (B) 50%
- (C) 69%
- (D) 30.8%
- (E) 56%
- (F) 36%
- (G) 44%

22. A wire $\frac{9}{10}$ of a metre long is cut into equal pieces, each $\frac{3}{40}$ of a metre long. How many whole pieces are obtained? [\[Fractions & Decimals\]](#)

- (A) 3 pieces
- (B) 120 pieces
- (C) 9 pieces
- (D) 40 pieces
- (E) 6 pieces
- (F) 24 pieces
- (G) 12 pieces

23. Two cyclists start 120 km apart and ride straight toward each other, one at 18 km/h and the other at 22 km/h. After how many hours do they meet? [\[Motion & Time\]](#)

- (A) 4 h
- (B) 6 h
- (C) 3.33 h
- (D) 3 h
- (E) 5 h
- (F) 2.4 h
- (G) 1.5 h

24. In a triangle, angle A is twice angle B, and angle C is 30 degrees more than angle B. What is the size of angle B? [\[The Triangle & its Properties\]](#)

- (A) 37.5 degrees
- (B) 52.5 degrees
- (C) 45 degrees
- (D) 40 degrees
- (E) 35 degrees
- (F) 75 degrees
- (G) 30 degrees

25. Pancreatic juice digests starch. In a tube it breaks down 12 g of starch in 4 minutes at a steady rate. How long, in minutes and seconds, will it take to break down 19 g of starch? [\[Nutrition in Animals\]](#)

- (A) 7 min
- (B) 5 min 40 s
- (C) 6 min 3 s
- (D) 6 min 30 s
- (E) 6 min 20 s
- (F) 6 min 33 s
- (G) 6 min 19 s

26. Find the value of $1.2 \times 0.05 \times 0.4$. [\[Fractions & Decimals\]](#)

- (A) 0.048
- (B) 0.024
- (C) 0.24
- (D) 0.0024
- (E) 0.12
- (F) 0.006

27. A car's odometer reads 45,280 km at the start of a trip and 45,604 km at the end. The trip took 4 hours. What was the average speed? [\[Motion & Time\]](#)

- (A) 75 km/h
- (B) 162 km/h
- (C) 90 km/h
- (D) 40.5 km/h
- (E) 324 km/h
- (F) 81 km/h
- (G) 80 km/h

28. Two of the exterior angles of a triangle are 110 degrees and 130 degrees. What is the third interior angle of the triangle? [\[The Triangle & its Properties\]](#)

- (A) 40 degrees
- (B) 80 degrees
- (C) 50 degrees
- (D) 70 degrees
- (E) 120 degrees
- (F) 60 degrees
- (G) 30 degrees

29. Gastric juice in the stomach is acidic. A person's stomach holds 0.5 L of gastric juice containing 0.04 mole of hydrochloric acid in every litre. How many millimoles of acid are present in total? [\[Nutrition in Animals\]](#)

- (A) 40 mmol
- (B) 200 mmol
- (C) 2 mmol
- (D) 8 mmol
- (E) 20 mmol
- (F) 0.02 mmol

30. A fuel tank holds 48 litres. Right now it is $\frac{5}{8}$ full. How many more litres are needed to make it $\frac{11}{12}$ full? [\[Fractions & Decimals\]](#)

- (A) 11 L
- (B) 6 L
- (C) 8 L
- (D) 18 L
- (E) 30 L
- (F) 14 L

31. On a journey a body covers 40 m in the first 10 s, then rests for 5 s, then covers 30 m in the next 5 s. What is its average speed over the whole 20 seconds? [\[Motion & Time\]](#)

- (A) 7 m/s
- (B) 5 m/s
- (C) 4 m/s
- (D) 3.5 m/s
- (E) 3 m/s
- (F) 4.67 m/s
- (G) 2 m/s

32. A straight ladder 13 m long leans against a vertical wall, with its foot 5 m from the base of the wall. How high up the wall does the ladder reach? [\[The Triangle & its Properties\]](#)

- (A) 11 m
- (B) 18 m
- (C) 12 m
- (D) 12.5 m
- (E) 9 m
- (F) 144 m

33. A python eats prey equal to one-quarter of its body mass, then digests it over many days, absorbing one-eighth of the original meal each day. After 3 full days, what fraction of the original meal is still left?

[\[Nutrition in Animals\]](#)

- (A) $\frac{5}{8}$
- (B) $\frac{5}{24}$
- (C) $\frac{2}{8}$
- (D) $\frac{7}{8}$
- (E) $\frac{3}{4}$
- (F) $\frac{1}{2}$
- (G) $\frac{3}{8}$

34. Four numbers are 0.45, $\frac{7}{15}$, 0.4 and $\frac{11}{25}$. Which one is the largest? [\[Fractions & Decimals\]](#)

- (A) 0.45
- (B) $\frac{11}{25}$
- (C) they are all equal
- (D) 0.4
- (E) $\frac{7}{15}$
- (F) 0.44

- 35.** A wheel of diameter 70 cm rolls along the ground without slipping for 110 m. How many full turns does it make? (use $\pi = 22/7$) [\[Motion & Time\]](#)
- (A) 55 turns
 - (B) 157 turns
 - (C) 100 turns
 - (D) 50 turns
 - (E) 25 turns
 - (F) 35 turns
 - (G) 44 turns
- 36.** A median of a triangle is drawn from one vertex to the midpoint of the opposite side. If the whole triangle has an area of 48 cm^2 , what is the area of each of the two smaller triangles formed? [\[The Triangle & its Properties\]](#)
- (A) 48 cm^2
 - (B) 32 cm^2
 - (C) 8 cm^2
 - (D) 16 cm^2
 - (E) 96 cm^2
 - (F) 24 cm^2
- 37.** A cow's stomach has four chambers; the large rumen holds 64% of the total 80-litre capacity. How many litres do the other three chambers hold together? [\[Nutrition in Animals\]](#)
- (A) 25.6 L
 - (B) 32 L
 - (C) 16 L
 - (D) 48 L
 - (E) 28.8 L
 - (F) 36 L
 - (G) 51.2 L
- 38.** Find two-thirds of three-quarters of five-sixths of 144. [\[Fractions & Decimals\]](#)
- (A) 40
 - (B) 48
 - (C) 90
 - (D) 60
 - (E) 72
 - (F) 120
- 39.** A snail crawls at a steady 5 mm per second. How many metres does it travel in one hour? [\[Motion & Time\]](#)
- (A) 300 m
 - (B) 18 m
 - (C) 1.8 m
 - (D) 5 m
 - (E) 0.018 m
 - (F) 18000 m

- 40.** The three angles of a triangle are $(3x - 10)$ degrees, $(2x + 20)$ degrees and $(x + 50)$ degrees. Find the value of x . [\[The Triangle & its Properties\]](#)
- (A) 15
 - (B) 18
 - (C) 40
 - (D) 20
 - (E) 30
 - (F) 10
- 41.** A goat's daily feed is one-third leaves, one-quarter grain, and the rest hay. If the hay weighs 5 kg, what is the goat's total daily feed in kilograms? [\[Nutrition in Animals\]](#)
- (A) 7 kg
 - (B) 15 kg
 - (C) 12 kg
 - (D) 6 kg
 - (E) 10 kg
 - (F) 8.57 kg
 - (G) 17 kg
- 42.** A jug holds 1.5 litres of juice. Each glass takes 0.18 litres. After filling 7 glasses, how many litres of juice are left in the jug? [\[Fractions & Decimals\]](#)
- (A) 0.24 L
 - (B) 1.32 L
 - (C) 1.26 L
 - (D) 0.36 L
 - (E) 0.18 L
 - (F) 0.3 L
 - (G) 0.26 L
- 43.** A train 250 m long passes a standing pole completely in 10 seconds. What is the train's speed in km/h? [\[Motion & Time\]](#)
- (A) 100 km/h
 - (B) 2.5 km/h
 - (C) 72 km/h
 - (D) 25 km/h
 - (E) 36 km/h
 - (F) 90 km/h
 - (G) 250 km/h
- 44.** An equilateral triangle has a perimeter of 54 cm. What is the length of its altitude (height)? (use square root of 3 as about 1.732, round to two decimals) [\[The Triangle & its Properties\]](#)
- (A) 31.18 cm
 - (B) 15 cm
 - (C) 15.59 cm
 - (D) 18 cm
 - (E) 27 cm
 - (F) 12 cm

45. A 24-hour record of digestive juices shows: 1.5 L saliva, 2.5 L gastric juice, 0.5 L bile, 1.5 L pancreatic juice and 1.5 L intestinal juice. What fraction of the total is gastric juice (in lowest terms)? [\[Nutrition in Animals\]](#)
- (A) $\frac{5}{15}$
 - (B) $\frac{1}{4}$
 - (C) $\frac{1}{5}$
 - (D) $\frac{3}{10}$
 - (E) $\frac{1}{2}$
 - (F) $\frac{1}{3}$
 - (G) $\frac{2}{5}$
46. Express $\frac{3}{8} + 0.35$ as a single decimal. [\[Fractions & Decimals\]](#)
- (A) 0.725
 - (B) 7.25
 - (C) 0.685
 - (D) 0.375
 - (E) 0.65
 - (F) 0.72
 - (G) 0.71
47. The same train, 250 m long and travelling at 25 m/s, now crosses a bridge 150 m long. How many seconds does it take to cross the bridge completely? [\[Motion & Time\]](#)
- (A) 400 s
 - (B) 10 s
 - (C) 16 s
 - (D) 20 s
 - (E) 14 s
 - (F) 8 s
 - (G) 6 s
48. A right-angled triangle has two equal legs, each 8 cm long. What is the length of its hypotenuse? (use square root of 2 as about 1.414) [\[The Triangle & its Properties\]](#)
- (A) 5.66 cm
 - (B) 12.8 cm
 - (C) 11.3 cm
 - (D) 8 cm
 - (E) 11.31 cm
 - (F) 16 cm
 - (G) 128 cm
49. An adult's small intestine is about 6 m long, while a young child's is about 250 cm. By what percent is the child's small intestine shorter than the adult's? (round to two decimals) [\[Nutrition in Animals\]](#)
- (A) 58.33%
 - (B) 250%
 - (C) 60%
 - (D) 58%
 - (E) 41.67%
 - (F) 350%
 - (G) 140%

50. A certain number multiplied by 0.6 gives 0.45. What is the number? [\[Fractions & Decimals\]](#)

- (A) 7.5
- (B) 0.075
- (C) 0.75
- (D) 0.9
- (E) 0.85
- (F) 1.33

51. A scooter goes from town A to town B at 30 km/h and returns along the same road at 20 km/h. What is its average speed for the whole round trip? [\[Motion & Time\]](#)

- (A) 12 km/h
- (B) 28 km/h
- (C) 25 km/h
- (D) 26 km/h
- (E) 24 km/h
- (F) 22 km/h
- (G) 50 km/h

52. In triangle ABC, angle A is a right angle (90 degrees) and angle B is twice angle C. What is the size of angle B? [\[The Triangle & its Properties\]](#)

- (A) 70 degrees
- (B) 60 degrees
- (C) 50 degrees
- (D) 30 degrees
- (E) 40 degrees
- (F) 45 degrees
- (G) 90 degrees

53. A blue whale gulps 70,000 L of water in one mouthful and gulps 4 times every minute while feeding. How many litres of water does it filter in 15 minutes? [\[Nutrition in Animals\]](#)

- (A) 4,200,000 L
- (B) 17,500 L
- (C) 1,050,000 L
- (D) 4,800,000 L
- (E) 2,800,000 L
- (F) 280,000 L
- (G) 420,000 L

54. Add the four fractions $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20}$ and give the answer in lowest terms. [\[Fractions & Decimals\]](#)

- (A) $\frac{5}{6}$
- (B) $\frac{9}{10}$
- (C) $\frac{1}{5}$
- (D) $\frac{3}{4}$
- (E) $\frac{7}{10}$
- (F) $\frac{4}{9}$
- (G) $\frac{4}{5}$

- 55.** On a clock, the minute hand turns at a steady rate. Through how many degrees does the minute hand turn between 3:00 and 3:25? [\[Motion & Time\]](#)
- (A) 180 degrees
 - (B) 90 degrees
 - (C) 120 degrees
 - (D) 144 degrees
 - (E) 25 degrees
 - (F) 300 degrees
 - (G) 150 degrees
- 56.** In a triangle, one angle is x and another is $(x + 20)$ degrees. The exterior angle at the third vertex is 130 degrees. Find the value of x . [\[The Triangle & its Properties\]](#)
- (A) 55
 - (B) 50
 - (C) 40
 - (D) 65
 - (E) 45
 - (F) 75
 - (G) 35
- 57.** A glucose drip delivers fluid with 0.9 g of glucose in every 100 mL. A 500 mL bag is run slowly over 5 hours. How many grams of glucose enter the body per hour? [\[Nutrition in Animals\]](#)
- (A) 1.8 g
 - (B) 0.45 g
 - (C) 2.25 g
 - (D) 9 g
 - (E) 0.18 g
 - (F) 0.9 g
 - (G) 4.5 g
- 58.** A pen costs Rs 13.75. A shopkeeper has Rs 200. What is the greatest number of pens that can be bought, and how much money is left over? [\[Fractions & Decimals\]](#)
- (A) 14 pens, Rs 12.50 left
 - (B) 16 pens, nothing left
 - (C) 13 pens, Rs 21.25 left
 - (D) 15 pens, Rs 0 left
 - (E) 15 pens, Rs 7.50 left
 - (F) 14 pens, Rs 6.25 left
 - (G) 14 pens, Rs 7.50 left
- 59.** An object moves at a uniform 15 m/s for 8 s, then stops for 4 s, then moves at a uniform 25 m/s for 6 s. What total distance does it cover? [\[Motion & Time\]](#)
- (A) 240 m
 - (B) 300 m
 - (C) 150 m
 - (D) 330 m
 - (E) 120 m
 - (F) 210 m
 - (G) 270 m

60. A right-angled triangle has legs of 6 cm and 8 cm. What is the length of the median drawn to its hypotenuse? [\[The Triangle & its Properties\]](#)

- (A) 5 cm
- (B) 10 cm
- (C) 8 cm
- (D) 4 cm
- (E) 7 cm
- (F) 6 cm
- (G) 3 cm

61. When too much is eaten, the body stores extra energy as fat (9 kcal per gram). A person takes in 200 extra kcal every day for one week. How many grams of fat could this build up? (round to one decimal)

[\[Nutrition in Animals\]](#)

- (A) 1400 g
- (B) 140 g
- (C) 155.6 g
- (D) 22.2 g
- (E) 160 g
- (F) 177.8 g
- (G) 156 g

62. By how much is the larger of $\frac{5}{6}$ and 0.8 greater than the smaller? Give the answer as a fraction in lowest terms. [\[Fractions & Decimals\]](#)

- (A) $\frac{1}{10}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{30}$
- (D) $\frac{1}{5}$
- (E) $\frac{2}{15}$
- (F) $\frac{5}{30}$
- (G) $\frac{1}{15}$

63. A 400 m running track is completed by an athlete in 50 seconds at a steady pace. What is the athlete's speed in km/h? [\[Motion & Time\]](#)

- (A) 25.6 km/h
- (B) 32 km/h
- (C) 400 km/h
- (D) 28 km/h
- (E) 8 km/h
- (F) 36 km/h
- (G) 28.8 km/h

64. Can a triangle be formed with sides of length 5 cm, 9 cm and 14 cm? [\[The Triangle & its Properties\]](#)

- (A) Yes, because $9 + 14$ is greater than 5
- (B) No, because 5×9 is less than 14
- (C) Yes, it forms a right-angled triangle
- (D) No, because 14 is too long to be a side
- (E) Yes, because all sides are different
- (F) Yes, because $5 + 14$ is greater than 9
- (G) No, because $5 + 9$ is not greater than 14

65. Food moves through the whole food canal in about 30 hours. It spends roughly 4 hours in the stomach and 6 hours in the small intestine, and almost all the rest in the large intestine. What fraction of the journey is spent in the large intestine (lowest terms)? [\[Nutrition in Animals\]](#)

- (A) $10/30$
- (B) $2/3$
- (C) $1/3$
- (D) $2/5$
- (E) $1/5$
- (F) $3/5$

66. A rod 2.4 m long is painted in three parts: $1/3$ of it red, 0.25 of it blue, and the rest green. How long is the green part in metres? [\[Fractions & Decimals\]](#)

- (A) 1.6 m
- (B) 1.0 m
- (C) 1.4 m
- (D) 1.2 m
- (E) 0.6 m
- (F) 0.8 m
- (G) 0.4 m

67. During a storm you see a lightning flash and hear the thunder 5 seconds later. Taking the speed of sound as 340 m/s (and light as instant), how far away is the storm in kilometres? [\[Motion & Time\]](#)

- (A) 17 km
- (B) 2.0 km
- (C) 3.4 km
- (D) 1.7 km
- (E) 0.68 km
- (F) 68 km

68. A triangle has a base of 14 cm and an area of 84 cm^2 . If the base is doubled but the area is kept the same, what is the new height? [\[The Triangle & its Properties\]](#)

- (A) 7 cm
- (B) 6 cm
- (C) 168 cm
- (D) 24 cm
- (E) 3 cm
- (F) 84 cm

69. A leech feeds by sucking 15 mL of blood at one meal, and it feeds only once every 6 months. Over a period of 3 years, how many millilitres of blood does it take in total? [\[Nutrition in Animals\]](#)

- (A) 540 mL
- (B) 90 mL
- (C) 30 mL
- (D) 360 mL
- (E) 45 mL
- (F) 180 mL
- (G) 15 mL

70. Evaluate $(\frac{3}{5} \text{ divided by } \frac{9}{10})$ multiplied by $(\frac{4}{7} \text{ divided by } \frac{8}{21})$. [\[Fractions & Decimals\]](#)

- (A) $\frac{5}{6}$
- (B) 1
- (C) $\frac{2}{3}$
- (D) $\frac{4}{9}$
- (E) $\frac{9}{4}$
- (F) $\frac{3}{2}$

71. A toy car covers 2 m in the 1st second, 4 m in the 2nd, 6 m in the 3rd and 8 m in the 4th second. What is its average speed over these 4 seconds? [\[Motion & Time\]](#)

- (A) 5 m/s
- (B) 2 m/s
- (C) 20 m/s
- (D) 8 m/s
- (E) 4 m/s
- (F) 6 m/s
- (G) 10 m/s

72. The angles of a triangle are in the ratio 1 : 1 : 2. What is the size of the largest angle? [\[The Triangle & its Properties\]](#)

- (A) 120 degrees
- (B) 100 degrees
- (C) 72 degrees
- (D) 60 degrees
- (E) 90 degrees
- (F) 45 degrees

73. Microbes in a cow's gut ferment cellulose from grass. They ferment 18 kg of cellulose and make 2.5 L of gas per kilogram. If this gas is released steadily over 24 hours, how many litres are released per hour (round to three decimals)? [\[Nutrition in Animals\]](#)

- (A) 3.75 L
- (B) 0.533 L
- (C) 2.5 L
- (D) 1.875 L
- (E) 18.75 L
- (F) 45 L

74. A worker finishes $\frac{2}{9}$ of a job on Monday, $\frac{1}{6}$ on Tuesday and $\frac{5}{18}$ on Wednesday. What fraction of the job is still left after Wednesday? [\[Fractions & Decimals\]](#)

- (A) $\frac{11}{18}$
- (B) $\frac{5}{18}$
- (C) $\frac{2}{3}$
- (D) $\frac{7}{18}$
- (E) $\frac{1}{6}$
- (F) $\frac{1}{2}$
- (G) $\frac{1}{3}$

75. Pendulum A has a time period of 2 s. Pendulum B swings 1.5 times as fast as A (its frequency is 1.5 times A's). What is pendulum B's time period? [\[Motion & Time\]](#)

- (A) 4 s
- (B) 3 s
- (C) 1.5 s
- (D) 0.67 s
- (E) 2 s
- (F) 0.75 s
- (G) 1.33 s

76. An isosceles triangle has side lengths of 5 cm and 11 cm (one of these is repeated). What is the perimeter of the triangle? [\[The Triangle & its Properties\]](#)

- (A) 11 cm
- (B) 22 cm
- (C) 15 cm
- (D) 21 cm
- (E) 33 cm
- (F) 16 cm
- (G) 27 cm

77. A baby drinks 150 mL of milk six times a day. The milk is 7% lactose (milk sugar) by mass, and 1 mL of milk weighs about 1 g. How many grams of lactose does the baby take in per day? [\[Nutrition in Animals\]](#)

- (A) 9 g
- (B) 6.3 g
- (C) 10.5 g
- (D) 900 g
- (E) 42 g
- (F) 70 g
- (G) 63 g

78. Find the value of 0.2 cubed plus 0.3 squared. [\[Fractions & Decimals\]](#)

- (A) 0.098
- (B) 0.038
- (C) 0.29
- (D) 0.98
- (E) 0.0098
- (F) 0.108

79. A bus travels at 54 km/h for 20 minutes, then at 72 km/h for 15 minutes. What total distance does it cover? [\[Motion & Time\]](#)

- (A) 63 km
- (B) 30 km
- (C) 54 km
- (D) 42 km
- (E) 36 km
- (F) 18 km
- (G) 45 km

- 80.** The three exterior angles of a triangle are $(2x)$, $(3x)$ and $(4x)$ degrees. What is the size of the smallest exterior angle? [\[The Triangle & its Properties\]](#)
- (A) 72 degrees
 - (B) 90 degrees
 - (C) 100 degrees
 - (D) 120 degrees
 - (E) 80 degrees
 - (F) 40 degrees
- 81.** The tongue carries about 10,000 taste buds, and they are replaced about once every 2 weeks. About how many taste buds are replaced per day (to the nearest whole number)? [\[Nutrition in Animals\]](#)
- (A) 1428 buds
 - (B) 714 buds
 - (C) 71 buds
 - (D) 5000 buds
 - (E) 700 buds
 - (F) 1000 buds
 - (G) 357 buds
- 82.** A chocolate bar is broken into 8 equal squares. Riya eats 3 of the squares and Sam eats one-quarter of the whole bar. What fraction of the bar is left over? [\[Fractions & Decimals\]](#)
- (A) $\frac{2}{8}$
 - (B) $\frac{5}{24}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{5}{8}$
 - (E) $\frac{1}{2}$
 - (F) $\frac{3}{8}$
 - (G) $\frac{3}{4}$
- 83.** Two trains, one 120 m long and one 180 m long, move directly toward each other at 15 m/s and 10 m/s. How long do they take to completely pass each other? [\[Motion & Time\]](#)
- (A) 12 s
 - (B) 20 s
 - (C) 15 s
 - (D) 10 s
 - (E) 30 s
 - (F) 6 s
- 84.** A triangle has sides of 8 cm, 15 cm and 17 cm. After checking it is right-angled, find its area. [\[The Triangle & its Properties\]](#)
- (A) 60 cm^2
 - (B) 127.5 cm^2
 - (C) 64 cm^2
 - (D) 120 cm^2
 - (E) 68 cm^2
 - (F) 30 cm^2

- 85.** Cattle teeth are of four kinds in the ratio incisors : canines : premolars : molars = 2 : 1 : 2 : 3, and there are 32 teeth in all. How many molars are there? [\[Nutrition in Animals\]](#)
- (A) 6 molars
 - (B) 9 molars
 - (C) 16 molars
 - (D) 12 molars
 - (E) 4 molars
 - (F) 8 molars
 - (G) 3 molars
- 86.** A cake weighs 1.6 kg. Seven-eighths of it is served, cut into equal slices of 0.07 kg each. How many full slices are served? [\[Fractions & Decimals\]](#)
- (A) 16 slices
 - (B) 18 slices
 - (C) 14 slices
 - (D) 22 slices
 - (E) 23 slices
 - (F) 200 slices
 - (G) 20 slices
- 87.** A 'seconds pendulum' has a time period of exactly 2 seconds. How many complete swings does it make in 5 minutes? [\[Motion & Time\]](#)
- (A) 75 swings
 - (B) 100 swings
 - (C) 120 swings
 - (D) 600 swings
 - (E) 300 swings
 - (F) 250 swings
 - (G) 150 swings
- 88.** In triangle PQR, angle P is 64 degrees, and angle Q is 18 degrees more than angle R. What is the size of angle Q? [\[The Triangle & its Properties\]](#)
- (A) 63 degrees
 - (B) 67 degrees
 - (C) 72 degrees
 - (D) 49 degrees
 - (E) 80 degrees
 - (F) 116 degrees
- 89.** In the large intestine, water is absorbed from the leftover food. 800 mL of watery waste enters, and 92.5% of it is absorbed back into the body. How many millilitres are left to be passed out? [\[Nutrition in Animals\]](#)
- (A) 60 mL
 - (B) 80 mL
 - (C) 92.5 mL
 - (D) 600 mL
 - (E) 120 mL
 - (F) 7.5 mL

90. Evaluate $4\frac{2}{5}$ minus 1 and $\frac{7}{10}$ plus 0.3. [\[Fractions & Decimals\]](#)

- (A) 3.3
- (B) 2.4
- (C) 2.7
- (D) 3.0
- (E) 1.0
- (F) 4.4
- (G) 3.6

91. A cheetah sprints at 112 km/h. What is this speed in metres per second (round to two decimals)? [\[Motion & Time\]](#)

- (A) 28 m/s
- (B) 62.22 m/s
- (C) 0.031 m/s
- (D) 403.2 m/s
- (E) 31.1 m/s
- (F) 112 m/s
- (G) 31.11 m/s

92. A rectangular field measures 24 m by 7 m. A straight path is laid along its diagonal. How long is the diagonal path? [\[The Triangle & its Properties\]](#)

- (A) 23 m
- (B) 20 m
- (C) 12.5 m
- (D) 25 m
- (E) 576 m
- (F) 17 m
- (G) 31 m

93. To test five saliva-treated food samples for starch, each test uses 2 drops of iodine solution. The dropper delivers 20 drops per millilitre. How many millilitres of iodine are used for all five tests? [\[Nutrition in Animals\]](#)

- (A) 5 mL
- (B) 1 mL
- (C) 0.5 mL
- (D) 10 mL
- (E) 0.1 mL
- (F) 0.25 mL
- (G) 2 mL

94. A 0.75 L bottle of squash is 64% full. The squash is poured into cups of 0.05 L each. How many full cups can be poured, and how much squash is left over? [\[Fractions & Decimals\]](#)

- (A) 9 cups, 50 mL left
- (B) 12 cups, 0 mL left
- (C) 8 cups, 80 mL left
- (D) 9 cups, 20 mL left
- (E) 10 cups, 30 mL left
- (F) 9 cups, 0 mL left
- (G) 9 cups, 30 mL left

- 95.** A bus's odometer reads 12,450 km and later 12,810 km after a steady-speed journey at an average of 60 km/h. How long did the journey take? [\[Motion & Time\]](#)
- (A) 7.2 h
 - (B) 3 h
 - (C) 12 h
 - (D) 5 h
 - (E) 4 h
 - (F) 6 h
- 96.** An isosceles right-angled triangle has a hypotenuse of length 10 times the square root of 2 cm. What is the length of each equal leg? [\[The Triangle & its Properties\]](#)
- (A) 20 cm
 - (B) 7.07 cm
 - (C) 14.14 cm
 - (D) 10 cm
 - (E) 10 times root 2 cm
 - (F) 5 cm
 - (G) 100 cm
- 97.** An adult human secretes about 1.5 L of saliva each day. If this saliva flows evenly over the 20 hours a person is awake, how many millilitres of saliva flow per minute (round to two decimals)? [\[Nutrition in Animals\]](#)
- (A) 2.5 mL
 - (B) 1.5 mL
 - (C) 1.25 mL
 - (D) 12.5 mL
 - (E) 0.075 mL
 - (F) 0.0125 mL
- 98.** Find the average (mean) of the four decimals 2.5, 3.75, 1.25 and 4.5. [\[Fractions & Decimals\]](#)
- (A) 4.0
 - (B) 3.25
 - (C) 3.0
 - (D) 12
 - (E) 2.5
 - (F) 2.75
- 99.** A boy walks to school at 4 km/h and the trip takes 30 minutes. He walks back home along the exact same route at 6 km/h. How many minutes does the return journey take? [\[Motion & Time\]](#)
- (A) 20 min
 - (B) 12 min
 - (C) 24 min
 - (D) 40 min
 - (E) 30 min
 - (F) 15 min

100. In a triangle, the largest angle is 5 times the smallest angle, and the middle angle is 4 times the smallest. What is the size of the middle angle? [\[The Triangle & its Properties\]](#)

- (A) 54 degrees
- (B) 72 degrees
- (C) 90 degrees
- (D) 18 degrees
- (E) 100 degrees
- (F) 80 degrees